

Evolutionary Biology: Origin, Mechanisms, and Evidence

Evolutionary biology is the study of the history of life forms on Earth. To comprehend the changes in flora and fauna that have occurred over millions of years, it is necessary to examine the origin of life within the broader context of the evolution of the universe, stars, and the Earth itself. The universe is approximately 13.8 billion years old, while the Earth is believed to have formed about 4.5 billion years ago within the solar system of the Milky Way galaxy.

I. The Origin of Life and Early Earth

The Big Bang Theory and Early Conditions

The Big Bang Theory describes a singular, massive explosion of very dense matter that caused the universe to expand and its temperature to fall. Eventually, hydrogen and helium formed and condensed due to gravitation to form galaxies. On early Earth (4.5 billion years ago), there was no atmosphere. The surface was covered by water vapor, methane, carbon dioxide, and ammonia released from molten mass. UV rays from the sun decomposed water into hydrogen and oxygen; the lighter hydrogen escaped into space, while oxygen combined with ammonia and methane to form water and CO_2 . As the Earth cooled, water vapor fell as rain, filling depressions to form oceans. Life appeared approximately 500 million years after the Earth's formation (about 4 billion years ago).

Theories of the Origin of Life

- **Panspermia:** Early Greek thinkers believed units of life called "spores" were transferred to different planets, including Earth.
- **Spontaneous Generation:** The belief that life arose from decaying and rotting matter (e.g., straw or mud). This was disproved by **Louis Pasteur**, who used swan-neck flasks to demonstrate that life comes only from pre-existing life.
- **Chemical Evolution (Oparin-Haldane Theory):** Proposed that the first forms of life originated from pre-existing non-living organic molecules (RNA, proteins, etc.). This theory suggests that chemical evolution—the formation of diverse organic molecules from inorganic constituents—preceded the appearance of life.

S.L. Miller's Experiment (1953)

S.L. Miller provided experimental evidence for chemical evolution by simulating primitive Earth conditions in a laboratory.

- **Conditions:** High temperature (800°C), electric discharge in a closed flask, and a reducing atmosphere containing CH_4 , H_2 , NH_3 , and water vapor.
- **Result:** He observed the formation of **amino acids**. Similar experiments later produced sugars, nitrogen bases, pigments, and fats.

II. Mechanisms and Theories of Evolution

Darwinian Theory: Natural Selection

Charles Darwin, based on observations during his voyage on the *H.M.S. Beagle*, concluded that existing life forms share varying degrees of similarity with each other and with extinct forms.

- **Natural Selection:** Individuals with characteristics that allow better adaptability to their environment (climate, food, etc.) survive and leave more progeny.
- **Fitness:** Darwin defined fitness specifically as **reproductive fitness**.
- **Common Descent:** All existing life forms share common ancestors who lived at different periods in Earth's history.

Lamarckism vs. Mutation Theory

Theory, Proponent, Core Concept

Inheritance of Acquired Characters, Lamarck, "Evolution occurs via the "use and disuse" of organs (e.g., giraffes stretching necks)."

Mutation Theory, Hugo de Vries, "Evolution is caused by sudden, random, directionless mutations rather than minor variations."

Hugo de Vries coined the term **saltation** to describe a single-step large mutation that leads to speciation. Unlike Darwin's view of slow, gradual evolution, de Vries believed evolution was jumpy and sudden.

III. Evidence for Evolution

Paleontological Evidence

Fossils are the remains of hard parts of life forms found in sedimentary rock layers. A study of these layers indicates the geological period in which organisms existed. Fossils show that life forms have changed over time and that certain organisms are restricted to specific geological time spans.

Comparative Anatomy and Morphology

- **Divergent Evolution (Homology):** Different animals share similar anatomical structures that perform different functions due to adaptation to different needs. This indicates **common ancestry**.
- **Examples:** Forelimbs of whales, bats, cheetahs, and humans; vertebrate hearts/brains; thorns of *Bougainvillea* and tendrils of *Cucurbita*.
- **Convergent Evolution (Analogy):** Different structures evolve for the same function in similar habitats, resulting in outward similarity despite different origins.
- **Examples:** Wings of butterflies and birds; eyes of octopuses and mammals; flippers of penguins and dolphins; sweet potato (root) and potato (stem).

Anthropogenic Evolution

Evolution can occur rapidly due to human action. The use of herbicides, pesticides, and antibiotics has led to the selection of resistant varieties of pests and microbes in a timeframe of months or years rather than centuries.

IV. The Hardy-Weinberg Principle

This principle states that allele frequencies in a population are stable and constant from generation to generation in the absence of evolutionary forces. This state is called **genetic equilibrium**.

The Algebraic Equation: $(p + q)^2 = p^2 + 2pq + q^2 = 1$

- P : Frequency of the dominant allele.
- q : Frequency of the recessive allele.
- p^2 : Frequency of homozygous dominant individuals.
- q^2 : Frequency of homozygous recessive individuals.
- $2pq$: Frequency of heterozygotes.

Factors Disturbing Equilibrium:

1. **Gene Migration/Flow:** Movement of genes into or out of a population.
2. **Genetic Drift:** Random changes in allele frequency by chance (includes the **Founder Effect**).
3. **Mutation:** Spontaneous heritable changes in DNA.
4. **Genetic Recombination:** Reshuffling of genes during gamete formation.
5. **Natural Selection:** Differential survival and reproduction.

V. Evolution Quiz

Instructions: Answer the following questions in 2-3 sentences based on the source text.

1. What does the Big Bang Theory attempt to explain regarding the origin of the universe?
2. How did the first oceans form on Earth according to the geological account?
3. Explain the concept of "Panspermia."
4. How did Louis Pasteur disprove the theory of spontaneous generation?
5. What were the specific conditions created by S.L. Miller in his 1953 experiment?
6. Define "Natural Selection" as understood by Charles Darwin.
7. What is the difference between homologous and analogous organs?
8. Describe "Adaptive Radiation" using Darwin's finches as an example.
9. What is "Saltation" according to Hugo de Vries?
10. How does the "Founder Effect" contribute to the formation of new species?

VI. Quiz Answer Key

1. **Big Bang Theory:** It explains the origin of the universe as a singular, massive explosion of dense matter from a point of singularity. This event caused the universe to expand and the temperature to drop, eventually leading to the formation of hydrogen, helium, and galaxies.
2. **Formation of Oceans:** As the early Earth cooled, water vapor released from molten mass condensed and fell as rain. This water filled all the depressions on the Earth's surface, eventually forming the oceans.
3. **Panspermia:** This was a theory favored by early Greek thinkers and some modern astronomers. It suggests that units of life called "spores" were brought to Earth and other planets from outer space.
4. **Louis Pasteur's Experiment:** Pasteur used pre-sterilized swan-neck flasks to show that life did not arise from killed yeast when the flask was sealed. However, when a flask

was open to the air, new living organisms arose, proving that life only comes from pre-existing life.

5. **Miller's Experimental Conditions:** Miller created an electric discharge in a closed flask at 800°C to simulate primitive Earth. The flask contained a reducing atmosphere of methane (CH_4), hydrogen (H_2), ammonia (NH_3), and water vapor.
6. **Natural Selection:** It is the process where individuals with hereditary variations that allow better adaptability to their environment survive and outbreed others. Darwin defined this "fitness" primarily as reproductive fitness, meaning the fittest leave more progeny.
7. **Homology vs. Analogy:** Homologous organs share a similar anatomical structure and common ancestry but perform different functions (divergent evolution). Analogous organs perform similar functions but have different anatomical origins and do not imply common ancestry (convergent evolution).
8. **Adaptive Radiation:** This is the evolution of different species in a given geographical area starting from a common point. Darwin's finches evolved from a single seed-eating ancestor into various forms with different beaks to exploit new food sources like insects or vegetation.
9. **Saltation:** Coined by Hugo de Vries, saltation refers to a single-step, large-scale mutation. He believed these sudden, random mutations were the primary cause of speciation, rather than the minor, gradual variations suggested by Darwin.
10. **Founder Effect:** This occurs when a small group of individuals drifts away from the original population to start a new one. The allelic frequency in this new sample can become so different that they form a new species, with the original drifted population acting as the "founders."

VII. Essay Questions

Instructions: Use the source context to develop comprehensive responses for the following topics.

1. Compare and contrast the views of Charles Darwin and Hugo de Vries regarding the mechanism of evolution and the nature of variations.
2. Describe the chemical evolution theory proposed by Oparin and Haldane and explain how S.L. Miller provided experimental validation for this theory.
3. Discuss the evidence for evolution provided by comparative anatomy, clearly distinguishing between divergent and convergent evolution with examples from the text.
4. Trace the evolution of man from *Dryopithecus* to modern *Homo sapiens*, noting changes in brain capacity, diet, and cultural developments.
5. Explain the Hardy-Weinberg Principle and detail the five factors that can disrupt genetic equilibrium in a population.

VIII. Glossary of Key Terms

- **Adaptive Radiation:** The process of evolution of different species in a given geographical area starting from a common ancestral point and radiating to other geographical areas.

- **Allele Frequency:** The proportion of a specific allele relative to the total alleles for a gene in a population's gene pool.
- **Anthropogenic Action:** Human-induced activities, such as the use of pesticides or antibiotics, that cause rapid evolutionary changes in organisms.
- **Biogenesis:** The principle that living organisms can only arise from other pre-existing living organisms.
- **Coelacanth:** A lobe-finned fish discovered in South Africa in 1938, believed to be the ancestor of the first amphibians.
- **Founder Effect:** A drastic change in allele frequency that occurs when a small group of individuals becomes isolated from the original population to form a new species.
- **Genetic Drift:** A change in gene frequency that occurs in a population due to random events or chance rather than natural selection.
- **Genetic Equilibrium:** A state in which the total gene pool (genes and their alleles) remains constant in a population over generations.
- **Hominid:** A member of the family that includes humans and their fossil ancestors; *Homo habilis* was the first human-like hominid.
- **Mutation:** A sudden, random, and directionless heritable change in the DNA sequence of an organism.
- **Paleontology:** The study of the history of life on Earth through the examination of plant and animal fossils.
- **Stochastic Process:** A process based on random events and chance, such as the occurrences in nature and mutations that drive evolution.